**Science · GROW · Autumn 2 · Week 1 · GROW_SCI_A2_W01_P1**

# TA Brief

I can explain how Earth's rotation causes day, night, and time zones, and how Earth's tilted axis causes changing day length and seasons.

**At check-out, record E for explaining day and night against the pupil's target.**

**At check-out, record E for today's lesson against each pupil's target on the evidence app.**

**No cold calling:** Invite a response, give thinking time, and accept a pass. Wait, point to Reference A or B, rephrase, then offer a two-word choice.

**During You do:** Record actual **I · S · C** beside each task. Use the optional participation-loop prompt inside You do so the pupil responds to feedback with an adult, and mark **R** only after the pupil has responded.

**Feedback codes** VF · WS · I · NS + … · E · R · // · ?
Give one feedback code. My response is the pupil's improvement to the work.
Adult checks the response and ticks it, or writes R (Pupil responded to feedback).

**Exit pupil voice**
What helped you today, and what would you change?
Accept a pointed, spoken, signed, drawn, written or chosen response. A pupil may pass.

## Stage timings

| Stage | Minutes |
|---|---|
| Today I can | 0 |
| Arrival | 5 |
| Starter | 4 |
| I do | 6 |
| We do | 7 |
| You do | 10 |
| Review and improve | 4 |
| Exit ticket | 4 |
| Tools & print | 0 |

**Total: 40 minutes**

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# Guidance and staff card

**Staff card · Science · GROW · Autumn 2 · Week 1 · GROW_SCI_A2_W01_P1**

**At check-out, record E for explaining day and night against the pupil's target.**

**At check-out, record E for today's lesson against each pupil's target on the evidence app.**

Invite a response. Give thinking time. Every pupil may pass or share privately. Never use cold calling.

**Return-to-learning route:** Agree a quiet pause when needed, offer the regulation words beside the colour (Green · ready to start, Amber · need a small adjustment, Red · need a quiet pause), then re-enter through one low-demand visual step (pointing to one position on Reference A or one marker on the Arrival drawing) before resuming the route task.

**Transitions and environment:** Use soft instrumental music for transitions if the class plan says so. Keep any demonstration lamp cool, stable, and aimed away from eyes; keep walkways clear.

**Optional participation-loop prompt (inside You do, not a separate stage):** The pupil responds to feedback with an adult, and R is marked only after the pupil has responded.

**Feedback codes** VF · WS · I · NS + … · E · R · // · ?
Give one feedback code. My response is the pupil’s improvement to the work.
Adult checks the response and ticks it, or writes R (Pupil responded to feedback).

## Feedback and marking codes

| Code | Meaning and staff use |
|---|---|
| VF | Verbal feedback given at this point |
| WS | Worked with adult support |
| I | Completed independently |
| NS + … | Next step: one specific, actionable improvement |
| E | Evidence captured on the evidence app |
| R | Responded: the pupil responds to feedback with an adult, and R is marked only after the pupil has responded |
| // | Pupil self-edit point, in the same colour on the same page |
| ? | Read this back to me (adult checks spoken meaning with the pupil) |

Keep the pupil's first attempt visible. Never require a special pen colour or box for corrections; edits happen in the same colour on the same page after //.

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# P17 · SLT planning and evaluation

**Lesson code:** GROW_SCI_A2_W01_P1 · **Arc:** introduce-explore (Lesson 1 of 2: Introduce and explore)

**Scheme reference:** GROW Science Scheme of Work · Autumn 2 Week 1 · Day and night (rebuilt from GROW W8A and live GROW W01 with Key Stage 3 progression on tilted axis, day length, seasons, and time zones).

**Evidence in this lesson:** Completed sequence and causal explanation tasks in the book or usual mode, with I · S · C support mark and one // review improvement. At check-out, record E for explaining day and night against the pupil's target. At check-out, record E for today's lesson against each pupil's target on the evidence app.

## Planned adaptations

| Area of need | Planned adaptation in this lesson | Where used |
|---|---|---|
| Cognition and learning | Fixed-left sunlight convention across all diagrams; chunked four-position North-Pole cycle (1 → 2 → 3 → 4) before Key Stage 3 tilt and time-zone extensions | Arrival, I do, We do inspector, Supported route |
| Communication and interaction | Point, sign, choice card, or adult exact-word scribing; optional participation-loop prompt with adult; adult read-back at Exit | Every stage check, You do, and Exit ticket |
| Social, emotional and mental health | Regulation check with words beside colour; private check-in; quiet pause and one-step return-to-learning route via Reference A | Arrival, You do, Review |
| Sensory and physical | Static high-contrast SVG diagrams with ≥18 px labels on normal-light pages (no blackout required); 44 px controls | All slides and Reference A/B |
| Language access | Explicit bridge between space viewpoint (rotation on a 23.5° tilted axis) and observer viewpoint (apparent movement and time zones) | I do word strip, Word help, Knowledge organiser |

## Where catch-up is embedded

1. **Arrival Q1** uses the self-contained Marker P / Marker Q picture cue so any pupil can start immediately regardless of prior attendance. 2. **I do** models all four quarter-turn positions step by step alongside the Key Stage 3 tilt and time-zone steps. 3. **Reference A and B** stay on the desk

throughout You do. 4. The **Catch-up sheet** is used for absence or access only, replacing a starting step rather than adding extra workload.

### If behind

Retain the Starter fixed-light check, the I do four-step model, the We do position check plus checkpoint question, one You do route explanation, and the Exit ticket. Trim a second We do round rather than rushing Review or Exit.

### Staff evaluation (complete after teaching)

| Evaluation prompt | Staff record |
|---|---|
| Delivered as planned | Yes / Partly / No — note any timing or route change |
| Success criteria met | Record which criteria were secure across Supported, Independent, Aspire, and Choice |
| Adaptations used and effect | Note which planned adaptations were used and how pupils responded |
| Coverage impact and next step | Ready for Week 1 Lesson 2 (Day and night: show what you know) or revisit rotation vs apparent motion in Arrival |
| Evidence banked | Book strip / work sample / At check-out, record E for explaining day and night against the pupil's target. |

**Made by Matt**

# P18 · Individual target opportunities

Select only the target that matches the pupil's existing individual plan held by the school. Do not copy personal details into this resource.

| Target domain | Observable opportunity in this lesson | Adult recording note |
|---|---|---|
| Cognition and learning | Orders Sunrise, Middle of the day, Sunset, and Night around the North-Pole model using Reference A | Record whether prompt was faded after Position 1 |
| Communication | Uses rotation, tilted axis, and apparent movement to explain why the Sun seems to cross our sky and why day length changes | Note mode used and whether adult read-back was confirmed |
| Regulation | Uses the regulation check or requests an adjustment and returns to one Reference A position | Follow agreed individual regulation plan |
| Sensory and physical | Uses the North-Pole rotation inspector or places printed position cards accurately without glare | Separate physical access support from science cueing |
| Independence | Completes the chosen route task with honest I, S, or C marking and optional participation loop | Mark actual independence per task; mark R only after pupil responds |

# Adult deployment and responsive feedback

**Two-adult deployment:** During I do and We do, one adult leads the North-Pole rotation model while the second adult monitors access and private responses. During You do, one adult supports pupils on the Supported/Choice routes while the other circulates across Independent and Aspire, swapping roles across the week so no pupil becomes prompt-dependent on one adult.

**Single-adult / cover deployment:** Keep Reference A displayed on screen during You do. Set Independent and Aspire learners started on Task 1, then check in with Supported/Choice learners using the two-option stems before rotating back for VF and NS + … during You do and Review.

**Responsive feedback and optional participation loop:** Inside You do (or during Review), use the optional participation-loop prompt: give private VF and one specific NS + …. The pupil responds to feedback with an adult (making a ***//*** edit in the same colour on the same page or explaining their reasoning orally), and **R** is marked only after the pupil has responded. At check-out, record E for explaining day and night against the pupil's target. At check-out, record E for today's lesson against each pupil's target on the evidence app.

**Made by Matt**

# First lesson back · once-only read

## 1. Five-minute book scan before teaching

Scan books for five minutes to check pupils' prior space and light vocabulary from Autumn 1 (Moon phases, reflected sunlight, orbit). Identify access barriers and actual starting points.

## 2. Existing NS + is history

Do not spend lesson time chasing historical feedback from previous weeks. Start fresh feedback from today's day-and-night rotation work.

## 3. Brief check with cover staff and TAs

Confirm that all diagrams keep the Sun fixed on the left and use the North-Pole anticlockwise view so staff prompts stay consistent.

## 4. Check the roll against evidence

If a pupil missed the start of the unit, give them Reference A and the Arrival picture cue; do not require them to complete missed sheets first.

## 5. One read after Exit, once

Read today's Exit ticket once against the three success criteria to set the starting route for Lesson 2 (Day and night: show what you know).

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# Teacher answers and topic guidance

Accept any scientifically equivalent wording in spoken, signed, pointed, drawn, or written form. Every question's wording is listed beside its number below, and the I do and We do model answers match the lesson page word for word.

## Starter

**1. It is night here. Has the Sun switched off? Choose the best scientific prediction to test with our fixed lamp and turning globe.**

Option A: The Sun stays shining all the time. Night happens at our marked place when Earth rotates on its axis so our place faces away from the Sun.

## I do model answer

**1. How does Earth's rotation on its tilted axis explain day and night, apparent Sun movement, time zones, length of day through the year, and seasons?**

Earth rotates anticlockwise (west to east) on its axis once every 24 hours, carrying a place through sunrise (top), middle of the day (left), sunset (bottom), and night (right). Because Earth turns 360° in 24 hours (15° per hour), time zones shift by 1 hour every 15° of longitude. Because Earth's axis is tilted at 23.5°, the hemisphere tilted towards the Sun has longer days and summer, while the hemisphere tilted away has shorter days and winter.

## We do model answer and checkpoint

**1. Match Positions 1 to 4 in the North-Pole view to the time of day and repair the faulty claim that the Sun switches off or orbits Earth daily.**

Position 1 (Top) is Sunrise (entering the sunlit left half). Position 2 (Left) is Middle of the day (facing the fixed Sun). Position 3 (Bottom) is Sunset (leaving the sunlit half). Position 4 (Right) is Night (facing away in the unlit half). Claim repair: the Sun stays shining, and night happens when Earth rotates on its axis once every 24 hours so our place faces away from the Sun.

**2. In the North-Pole model with the Sun fixed on the left, Marker X is at Position 3 (bottom boundary), turning anticlockwise out of the lit half into the unlit half. Which time of day is at Marker X, and why?**

Option A: Sunset, because Earth's anticlockwise rotation on its axis carries Marker X from the sunlit left half into the unlit right half.

## Arrival · Supported

1. **1. Look at the arrival picture cue. Which letter marks the place on the bright sunlit half of Earth?** — Marker P is on the bright sunlit half of Earth.
2. **2. Look at Marker Q on the shaded right half of the drawing. Why is no sunlight reaching Marker Q?** — Marker Q faces away from the Sun on the unlit half of Earth.
3. **3. Think back: what do we call the imaginary pole-to-pole line that Earth spins around?** — Earth spins around its axis.
4. **4. In a classroom model, what object does the fixed lamp represent?** — The fixed lamp represents the Sun.

## Arrival · Independent

1. **1. Use the arrival picture cue. Which marker is in darkness, and how does the drawing show this?** — Marker Q is in darkness because it is on the shaded right half of Earth, facing

away from the sunlight arrows.

2. **2. The Moon stays spherical all month. Why does the lit shape we see from Earth change during a month?** — The Sun always lights half of the Moon, and as the Moon orbits Earth our viewpoint shows different parts of that lit half.
3. **3. What is the difference between Earth's daily rotation and Earth's yearly orbit?** — Rotation is Earth spinning on its axis every 24 hours; orbit is Earth travelling around the Sun once a year.
4. **4. Why must we keep the model lamp still while turning the globe to investigate day and night?** — Keeping the lamp still isolates the effect of Earth's rotation so we can see how a turning planet causes day and night.

### Arrival · Aspire

1. **1. Use the arrival picture cue. As Earth rotates anticlockwise in this North-Pole view, what happens as a marker at the top turns into the left half?** — A marker at the top enters the sunlit left half, modelling sunrise followed by middle of the day.
2. **2. Earth rotates 360° in 24 hours. Through how many degrees of longitude does Earth turn in 1 hour, and how does this cause time zones?** — Earth turns through 15° per hour (360° ÷ 24 = 15°), so local solar time and time zones shift by 1 hour for every 15° of longitude.
3. **3. At what angle is Earth's axis tilted, and how does this tilted axis affect the length of day through the year?** — Earth's axis is tilted at 23.5°, which makes daylight hours longer when a hemisphere tilts towards the Sun and shorter when it tilts away.
4. **4. How does Earth's tilted axis explain why the Northern and Southern Hemispheres have opposite seasons?** — When the Northern Hemisphere tilts towards the Sun it gets longer days and more direct sunlight (summer), while the Southern Hemisphere tilts away and gets shorter days (winter).

### Arrival · Choice

1. **1. Look at the arrival picture cue. Point to the marker on the sunlit side of Earth.** — Marker P is on the sunlit side.
2. **2. Look at the arrival picture cue. Point to the marker on the dark shaded side of Earth.** — Marker Q is on the dark shaded side.
3. **3. Which body spins once every 24 hours to cause day and night?** — Earth spins on its axis once every 24 hours.
4. **4. Does the Sun switch off at night, or does it stay shining?** — The Sun stays shining all the time.

### You do · Supported mark scheme

1. **1. Use Reference A (North-Pole view with the fixed Sun on the left). Order the four daily moments starting at Position 1: Night, Sunset, Sunrise, Middle of the day.** — 1. Sunrise (entering the light) → 2. Middle of the day (facing the Sun) → 3. Sunset (leaving the light) → 4. Night (facing away). *(Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.)*
2. **2. Complete these sentences using the word bank (day, night, rotates): When our place faces the Sun it has ___. When our place faces away it has ___. Day turns to night because Earth ___.** — When our place faces the Sun it has day. When our place faces away

it has night. Day turns to night because Earth rotates. *(Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.)*

3. **3. Explain why darkness comes each evening without the Sun switching off.** — The Sun stays shining all the time, and darkness comes when Earth rotates on its axis so our place faces away from the Sun. *(Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.)*

## You do · Independent mark scheme

1. **1. Use Reference A. Track one marked place through Positions 1, 2, 3, and 4 in the North-Pole view. Name the time of day at each position and state the evidence clue.** — Position 1 (top): Sunrise, because the place enters the lit half. Position 2 (left): Middle of the day, because the place faces the Sun. Position 3 (bottom): Sunset, because the place leaves the lit half. Position 4 (right): Night, because the place faces away in the unlit half. *(Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.)*
2. **2. Explain why day turns to night using the words axis and rotation, then explain why the Sun appears to move across our sky from east to west.** — Earth spins on its axis once every 24 hours. This rotation carries each place into sunlight for day and away into shade for night. Because we rotate west to east with Earth, the Sun has an apparent movement from east to west across our sky. *(Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.)*
3. **3. Use Reference B to explain how Earth's 24-hour rotation causes time zones and why day length changes through the year.** — Earth rotates 360° in 24 hours (15° per hour), so time zones shift by 1 hour every 15° of longitude. Because Earth's axis is tilted at 23.5°, day length changes through the year as our hemisphere tilts towards or away from the Sun. *(Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.)*

## You do · Aspire mark scheme

1. **1. Explain how Earth's west-to-east rotation causes both the apparent east-to-west motion of the Sun and global time zones (using 360° ÷ 24 hours = 15° per hour).** — Earth rotates west to east on its axis once every 24 hours, making the Sun appear to move east to west. Because 360° ÷ 24 hours = 15° per hour, places 15° east enter sunlight 1 hour earlier, creating 1-hour time zone steps. *(Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.)*
2. **2. Explain how Earth's 23.5° tilted axis changes the length of day through the year between the summer solstice, winter solstice, and equinoxes.** — Because Earth's axis is tilted at 23.5°, a hemisphere tilted towards the Sun spends more than half of each 24-hour turn in sunlight (longer days in summer), while the hemisphere tilted away spends less than half in sunlight (shorter days in winter). At the equinoxes, day and night are about 12 hours everywhere. *(Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.)*
3. **3. Use the tilted axis (23.5°) and Earth's orbit to explain the seasons, and state why four equal quarter turns on a non-tilted model only represent an equinox.** — Seasons happen because Earth's 23.5° tilted axis keeps the same direction in space as Earth orbits the Sun, changing both day length and how high the Sun rises. A non-tilted four-quarter model shows equal 12-hour day and night, which only happens at the equinoxes. *(Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.)*

## You do · Choice mark scheme

1. **1. 1. In our North-Pole model with the Sun on the left, which half of Earth is in daylight: the left half or the right half? 2. What movement causes day and night: Earth's rotation or the Sun switching off?** — 1. Left half facing the Sun. 2. Earth's rotation on its axis. *(Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.)*
2. **2. 3. When the red marker moves from the lit half into the unlit half at the bottom (Position 3), is that Sunset or Sunrise? 4. How long does one full rotation take: 24 hours or 1 year?** — 3. Sunset. 4. 24 hours. *(Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.)*
3. **3. 5. Why does the Sun seem to cross our sky from east to west: Earth rotates west to east or the Sun orbits Earth daily? Add one short sentence using the word apparent.** — Earth rotates west to east. The Sun's daily path across the sky is an apparent movement caused by Earth spinning. *(Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.)*

## Exit ticket · Supported

1. **1. In the North-Pole model, Place X is on the unlit right half of Earth. Is Place X in day or night, and what does Earth do so Place X reaches daylight?** — Place X is in night. Earth rotates on its axis, carrying Place X around into the sunlit half.
2. **2. How many hours does one full rotation of Earth take?** — One full rotation of Earth takes about 24 hours.
3. **3. Why does the Sun not need to switch off for us to have night?** — The Sun stays shining all the time, and we have night when our place on Earth rotates away from the Sun.

## Exit ticket · Independent

1. **1. Explain why Place X on the unlit side of Earth has night and how it reaches sunrise and middle of the day.** — Place X faces away from the Sun so no direct sunlight reaches it. As Earth rotates anticlockwise (west to east), Place X turns into the lit half at sunrise and faces the Sun at middle of the day.
2. **2. Why does the Sun appear to move across our sky from east to west each day?** — We rotate from west to east with Earth, so our viewing direction sweeps past the Sun and makes it appear to move from east to west.
3. **3. How does Earth's 23.5° tilted axis change the length of day through the year?** — When our hemisphere tilts towards the Sun we stay in the lit half longer (longer summer days), and when it tilts away we stay in the lit half for less time (shorter winter days).

## Exit ticket · Aspire

1. **1. Explain Place X's night and later daylight using both the space viewpoint (rotation) and the Earth observer's viewpoint (apparent movement).** — From space, Earth rotates west to east on its axis, carrying Place X from the unlit half into sunlight. From Place X, the observer turns with Earth and sees the Sun appear to rise in the east and cross the sky to the west.
2. **2. How does Earth's 360° rotation in 24 hours (15° per hour) explain time zones?** — Because Earth rotates 15° of longitude every hour, places 15° further east turn into sunlight 1 hour earlier, so global time zones step by 1 hour every 15°.

3. **3. How does Earth's tilted axis (23.5°) explain changing length of day through the year and the seasons?** — Earth's 23.5° tilted axis means that as Earth orbits the Sun, the hemisphere tilted towards the Sun gets longer daylight hours and more direct sunlight (summer), while the hemisphere tilted away gets shorter daylight hours and lower sunlight (winter).

## Exit ticket · Choice

1. **1. Place X is on the shaded half facing away from the Sun. Choose: is Place X in day or in night?** — Place X is in night.
2. **2. What carries Place X from night into sunrise? Choose: Earth rotates on its axis / The Sun switches back on.** — Earth rotates on its axis.
3. **3. How long is one full rotation of Earth? Choose: 24 hours / 365 days.** — 24 hours.

## Exit pupil voice

**1. What helped you today, and what would you change?** — Accept any pointed, spoken, signed, drawn, written, or chosen reflection on what helped and what could change next time. A pupil may pass.

## Topic guidance and Key Stage 3 progression

- **Building on primary knowledge to reach Key Stage 3:** At primary level, pupils learn that Earth spins once every 24 hours to cause day and night. At Key Stage 3, connect this spin to **time zones** (360° ÷ 24 hours = 15° of longitude per hour) and connect Earth's **23.5° tilted axis** to changing **length of day through the year** and the **seasons**.
- **Does the Sun move around Earth each day?** Keep the model Sun fixed on the left and show that Earth rotates west to east on its axis once every 24 hours, creating the Sun's apparent east-to-west motion across the sky.
- **Tilted axis, day length, and seasons:** When the Northern Hemisphere tilts towards the Sun at the summer solstice, more than half of each Northern latitude circle is in sunlight (longer days and summer) while the Southern Hemisphere tilts away (shorter days and winter). At the equinoxes, day and night are about 12 hours worldwide.

**Made by Matt**

# Teacher and cover plan

**Lesson:** Science · GROW · Autumn 2 · Week 1 · GROW_SCI_A2_W01_P1 · **Title:** Day and night: introduce and explore · **Weekly arc:** Introduce and explore (Lesson 1 of 2)

**Learning goal:** I can explain how Earth's rotation causes day, night, and time zones, and how Earth's tilted axis causes changing day length and seasons.

**At check-out, record E for explaining day and night against the pupil's target.**

**At check-out, record E for today's lesson against each pupil's target on the evidence app.**

| Stage | Minutes | Pupil activity and teacher check |
|---|---|---|
| Today I can | 0 | Which half of Earth has day in the diagram, and what movement carries a marked place from day into night? |
| Arrival | 5 | Use Question 1 to check picture access first. Accept pointing, speech, sign, or book responses. |
| Starter | 4 | Every pupil commits before the reveal. Check that pupils distinguish the fixed light source from the rotating Earth. |
| I do | 6 | Build on primary knowledge to reach Key Stage 3: connect the 24-hour spin to 15°/h time zones and connect the 23.5° tilted axis to changing day length and seasons. |
| We do | 7 | Invite each pupil to predict and justify the position before clicking Check, then complete the We do checkpoint question before moving to You do. |
| You do | 10 | One adult supports the agreed access group; the other holds the main group, swapping across the week. Mark I, S, or C beside each task. Use the optional participation-loop prompt where helpful, marking R only after the pupil has responded. |
| Review and improve | 4 | Give private verbal feedback and one clear next step. Wait for the pupil response and |

| Stage | Minutes | Pupil activity and teacher check |
|---|---|---|
| | | engage with it before closing the loop. |
| Exit ticket | 4 | Receive each response and repeat the science back so the pupil can confirm or correct it. Use the exit to pick the starting point for Lesson 2. |
| Tools & print | 0 | Select only the sheet needed for access. Do not add catch-up work to a present pupil. |

**Total time:** 40 minutes. **Next lesson:** Day and night: show what you know.

**Check-out and participation loop:** Inside You do, use the optional participation-loop prompt (the pupil responds to feedback with an adult, and R is marked only after the pupil has responded). At check-out, record E for explaining how Earth's rotation causes day and night against the pupil's target.

**Equipment & safety:** Reference A and B sheets, route cards, optional cool LED torch, globe, and removable dot. Keep any torch beam aimed at the globe, never at eyes; normal-light diagrams are a full equivalent route.

**Made by Matt**